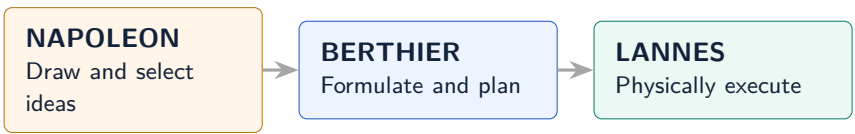
A THEORY OF INTELLIGENCE AND PRODUCTION

From Ideas to Action

Napoleon, Berthier, and Lannes



Research model

October 2026

The hypothesis

Intelligence is not a single factor of production.

NAPOLEON

Conception
What is worth trying?

Produces and evaluates a vague idea ξ.

BERTHIER

Mental execution
What plan implements it?

Formulates a problem and obtains a *fallible* solution $\hat{a}$.

LANNES

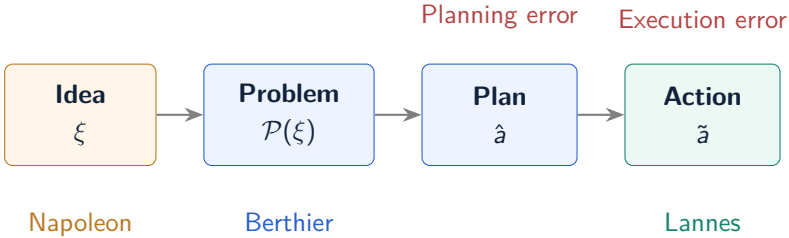
Physical execution
What actually happens?

Carries out the plan, possibly imperfectly: $\tilde{a}$.

Working hypothesis: current AI may disproportionately improve Berthier's capability, while an ASI would also be an exceptional Napoleon. Lannes remains necessary whenever output requires physical action.

The project pipeline

A vague idea is not yet a plan, and a plan is not yet an outcome.



An idea ξ specifies a desired possibility. Berthier converts it into a well-defined problem, finds a plan, and hands the plan to Lannes. The resulting physical action generates payoff Y.

Napoleon: idea search and selection

The first stage generates heterogeneity across potential projects.

Idea-generating technology

$$\xi_i \overset{\text{i.i.d.}}{\sim} F_A, \qquad \lambda_A = \frac{n_A}{\tau_A}.$$

- Each Napoleon draws one idea every τ_A units of time.
- n_A architects work in parallel.
- F_A describes the distribution of *ideas*, not merely the number generated.

Value of an idea

Let $v(\xi)$ be the value achievable by an optimal plan and perfect execution.

$$v(\xi) = \max_{a \in \mathcal{A}(\xi)} W(a; \xi).$$

Napoleon observes $v(\xi)$ in the baseline and decides whether the idea deserves downstream capacity.

Architectural progress can raise the arrival rate λ_A, shift the distribution of $v(\xi)$, or improve Napoleon's ability to recognize valuable ideas.

Berthier: turning an idea into a problem

A correct problem has an optimal plan, whether or not Berthier finds it.

The benchmark problem induced by an idea

$$\mathcal{P}(\xi): \qquad \max_{a \in \mathcal{A}(\xi)} W(a;\xi), \qquad a^*(\xi) \in \arg\max_{a \in \mathcal{A}(\xi)} W(a;\xi).$$

Formulation
Berthier translates a vague objective into decision variables, constraints, and payoffs.

Optimization
Berthier allocates scarce planning time τ_B to finding a good solution to that problem.

We first combine formulation and solution errors into one reduced-form deviation from $a^*(\xi)$; later the two can be distinguished.

Berthier: optimization mistakes

Even the right objective need not produce the best operational plan.

$$\boxed{\hat{a} = a^*(\xi) + \varepsilon_B} \qquad \mathbb{E}[\varepsilon_B \mid \xi] = 0, \quad \mathrm{Var}(\varepsilon_B \mid \xi) = \Omega_B.$$

What the error means
The planned location, timing, allocation, or sequence of orders differs from the best feasible solution.

What AI might change
A better Berthier produces $\Omega_B \downarrow$, may require less planning time $\tau_B \downarrow$, or both.

For additive errors, assume an interior optimum and locally feasible actions. A systematic formulation mistake can instead introduce a nonzero conditional bias.

Lannes: execution mistakes

Receiving a correct plan does not guarantee carrying it out.

$$\boxed{\tilde{a} = \hat{a} + \varepsilon_L} \qquad \mathbb{E}[\varepsilon_L \mid \xi, \varepsilon_B] = 0, \quad \mathrm{Var}(\varepsilon_L \mid \xi) = \Omega_L.$$

$$\underbrace{a^*(\xi)}_{\text{ideal plan}} \xrightarrow{+\varepsilon_B} \underbrace{\hat{a}}_{\text{Berthier's plan}} \xrightarrow{+\varepsilon_L} \underbrace{\tilde{a}}_{\text{Lannes's action}}$$

Examples: late arrival, miscommunication, local friction, inaccurate maneuvering, or inability to supply the commanded effort. One realization of ε_L may accidentally compensate for Berthier's mistake.

The model guarantees losses only in expectation, not for every realization.

A tractable payoff technology

Quadratic losses make the two kinds of mistakes separately interpretable.

$$W(a;\xi) = v(\xi) - \frac{1}{2}\big(a - a^*(\xi)\big)^\top H\big(a - a^*(\xi)\big)$$

- $H \succ 0$ measures how costly deviations from the ideal plan are.
- Berthier delivers $\hat{a} = a^*(\xi) + \varepsilon_B$.
- Lannes delivers $\tilde{a} = \hat{a} + \varepsilon_L$.
- Actual project output is $Y(\xi) = W(\tilde{a};\xi)$.

Take H and the error covariance matrices as independent of ξ in the baseline so that ideas can be ranked by $v(\xi)$ alone.

An additive expected-loss decomposition

Optimization mistakes and implementation mistakes reduce output separately.

$$\tilde{a} - a^*(\xi) = \varepsilon_B + \varepsilon_L.$$

Under mean-zero independent errors, the cross term vanishes:

$$\mathbb{E}[Y \mid \xi] = v(\xi) - \underbrace{\tfrac{1}{2}\operatorname{tr}(H\Omega_B)}_{\kappa_B} - \underbrace{\tfrac{1}{2}\operatorname{tr}(H\Omega_L)}_{\kappa_L}.$$

Berthier's loss

$$\kappa_B = \tfrac{1}{2}\operatorname{tr}(H\Omega_B)$$

An imperfect choice of plan.

Lannes's loss

$$\kappa_L = \tfrac{1}{2}\operatorname{tr}(H\Omega_L)$$

Imperfect implementation of that plan.

All three stages consume time

Different projects can be processed at different stages simultaneously.

Stage	Workers	Time per project	Rate
Napoleon: ideas	n_A	τ_A	$\lambda_A = n_A/\tau_A$
Berthier: plans	n_B	τ_B	$\mu_B = n_B/\tau_B$
Lannes: actions	n_L	τ_L	$\mu_L = n_L/\tau_L$

With a sufficient supply of accepted ideas, long-run throughput is constrained by

$$\boxed{K = \min\{\mu_B, \mu_L\}.}$$

Sequential latency of one project is $\tau_A + \tau_B + \tau_L$; long-run throughput can be faster because different projects overlap across stages.

Napoleon selects which ideas to pursue

Planning and execution slots should not be used on every draw.

Write $V = v(\xi)$ and let G_A denote its distribution. Assume V is observed by Napoleon.

Cutoff decision

$$\text{Accept idea} \iff V \geq z.$$

Ideas arrive at rate λ_A; an accepted idea occupies one slot at both later stages.

Acceptance flow

$$Q(z) = \lambda_A[1 - G_A(z)].$$

Capacity condition

$$Q(z) \leq K = \min\{\mu_B, \mu_L\}.$$

The opportunity cost of scarce downstream capacity gives Napoleon a reason to reject *positive-value* ideas when even better ones are available.

The optimal acceptance threshold

Take Berthier and Lannes to have idea-independent expected losses.

Define $\kappa = \kappa_B + \kappa_L$. The planner chooses a cutoff to maximize expected output per unit of time:

$$\begin{aligned} \max_{z} \quad & \mathcal{Y}(z) = \lambda_A \int_z^{\infty} (v - \kappa)\, dG_A(v) \\ \text{subject to} \quad & \lambda_A[1 - G_A(z)] \leq K. \end{aligned}$$

$$z^* = \max\left\{\kappa, \quad G_A^{-1}\left(1 - \min\left\{1, \frac{K}{\lambda_A}\right\}\right)\right\}.$$

Assumes a continuous value distribution and constant per-project stage times. The first term discards expected-loss projects; the second reserves scarce capacity for the best ideas.

AI improves Berthier along two margins

Faster plans and better plans are economically different technologies.

SPEED

$$\tau_B \downarrow \implies \mu_B \uparrow .$$

More ideas can proceed to execution *if Berthier is the binding bottleneck*.

ACCURACY

$$\Omega_B \downarrow \implies \kappa_B \downarrow .$$

Even if Lannes is already binding, each implemented project can generate more expected output.

Envelope result (away from threshold ties): $\dfrac{\partial \mathcal{Y}^*}{\partial \kappa_B} = -Q^*$. Reducing Berthier's expected loss by one unit raises value by the flow of accepted projects.

A numerical illustration

Lannes can bind while better planning continues to create value.

Scenario	Idea flow λ_A	Plan cap. μ_B	Action cap. μ_L	Output/day
Baseline	6	3	2	11.67
Better Berthier (AI)	6	8	2	14.17
Napoleon + Berthier (ASI)	9	8	2	22.39

Baseline: $V \sim U[0, 10]$, $\kappa_B = 1.5$, $\kappa_L = 1$. AI: $\kappa_B = 0.25$.
ASI: also improve architectural draws to $V \sim U[0, 14]$ and $\lambda_A = 9$.

Lannes still completes only **two projects per day**. Improvement occurs through selecting and executing *more valuable* projects with fewer planning errors. Purely illustrative parameter values, not a quantitative forecast. The top K/λ_A fraction of draws is accepted.

A natural extension: time buys accuracy

Planning time and physical execution time may influence error probabilities.

Suppose, for one project, more time lowers each stage's expected loss:

$$\kappa_B(t_B) = \frac{b}{t_B}, \qquad \kappa_L(t_L) = \frac{\ell}{t_L}, \qquad b, \ell > 0.$$

With a fixed combined time budget $T = t_B + t_L$, choose the division of labor:

$$\min_{t_B, t_L > 0} \frac{b}{t_B} + \frac{\ell}{t_L} \quad \text{s.t.} \quad t_B + t_L = T.$$

$$\boxed{\frac{t_B^*}{t_L^*} = \sqrt{\frac{b}{\ell}}.}$$

An AI improvement that lowers b reduces the planning time needed for a given accuracy. In the full pipeline, optimal time also depends on relative stage capacity and the value of accepted projects.

Lannes supplies a benchmark—but not perfect identification

The physical stage anchors ideas and plans to observed consequences.

What can be observed?

- Berthier's submitted plan $\hat{a}$.
- Lannes's realized action $\tilde{a}$.
- The realized payoff Y.

What can be learned?

$\tilde{a} - \hat{a}$ measures execution deviation.

$\hat{a} - a^*(\xi)$ measures planning deviation *if a credible benchmark optimum can be established.*

Repeated outcomes help test whether an architect's ideas were good.

Observing physical outcomes does not, by itself, separate poor ideas, poor plans, and execution noise. Benchmarks or additional variation are needed.

AI versus ASI

The model separates architectural from executive intelligence.

AI: the Berthier margin

- Faster planning: $\tau_B \downarrow$.
- Better planning: $\Omega_B \downarrow$.
- Same distribution of architectural ideas F_A.

ASI: also the Napoleon margin

- Generates more ideas: $\tau_A \downarrow$.
- Generates better ideas: F_A shifts upward in value.
- Retains strong planning: $\tau_B, \Omega_B \downarrow$.

Main tension: as long as Lannes is required, powerful cognitive intelligence faces a physical throughput bound. But idea quality and planning accuracy still matter *within* that bound.

Three questions for the theory

This is a starting point, not yet a complete general-equilibrium model.

1. **Endogenous complexity.** Do Napoleon's most ambitious ideas require more Berthier time, more Lannes time, or create larger error variances?

$$\tau_B(\xi),\ \tau_L(\xi),\ \Omega_B(\xi),\ \Omega_L(\xi).$$

2. **Occupational choices and wages.** How does cheaper Berthier labor change demand for Napoleon-like and Lannes-like workers?
3. **Learning and evaluation.** Can feedback from actual executions improve idea selection and distinguish failures of conception from failures of implementation?

Central question: does AI's first-order economic effect come from generating better ideas, or from making the existing stock of human ideas much easier to realize?

Appendix: closed form with uniform idea values

An explicit solution for idea selection and aggregate expected output.

Assume $V \sim U[0, \bar{v}]$, constant expected losses $\kappa = \kappa_B + \kappa_L$, and downstream capacity K.

$$z^* = \max\left\{\kappa,\ \bar{v}\left(1 - \min\left\{1, \frac{K}{\lambda_A}\right\}\right)\right\}.$$

The accepted flow and its average potential value are

$$Q^* = \lambda_A\left(1 - \frac{z^*}{\bar{v}}\right), \qquad \mathbb{E}[V \mid V \geq z^*] = \frac{\bar{v} + z^*}{2}.$$

Thus the optimized output rate is

$$\boxed{\mathcal{Y}^* = Q^*\left(\frac{\bar{v} + z^*}{2} - \kappa\right).}$$

Appendix: beyond independent, unbiased errors

The additive loss decomposition depends on specific assumptions.

Let $\mathbb{E}[\varepsilon_B] = b_B$, $\mathbb{E}[\varepsilon_L] = b_L$ and let $C_{BL} = \mathrm{Cov}(\varepsilon_B, \varepsilon_L)$.

$$\mathbb{E}[Y \mid \xi] = v(\xi) - \frac{1}{2}(b_B + b_L)^\top H(b_B + b_L) - \frac{1}{2}\,\mathrm{tr}\big[H(\Omega_B + \Omega_L + C_{BL} + C_{BL}^\top)\big].$$

- **Biased planning:** Berthier may systematically misunderstand Napoleon's idea.
- **Correlated error:** Lannes may correct, amplify, or respond to Berthier's mistakes.
- **Uncertain outcomes:** additional payoff shocks can be included without changing expected output if mean-zero and independent.